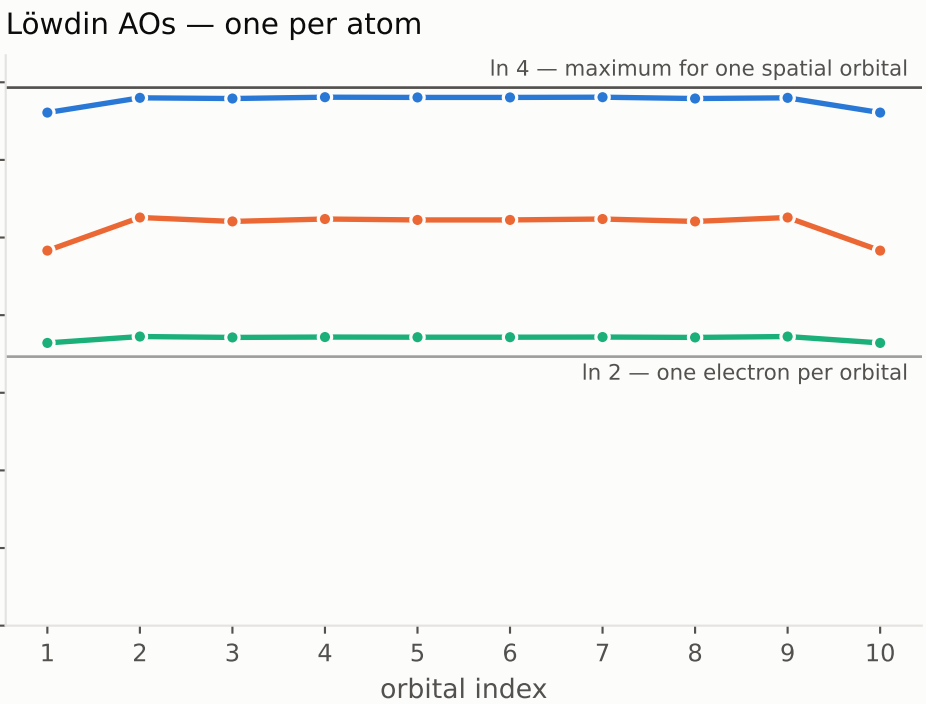
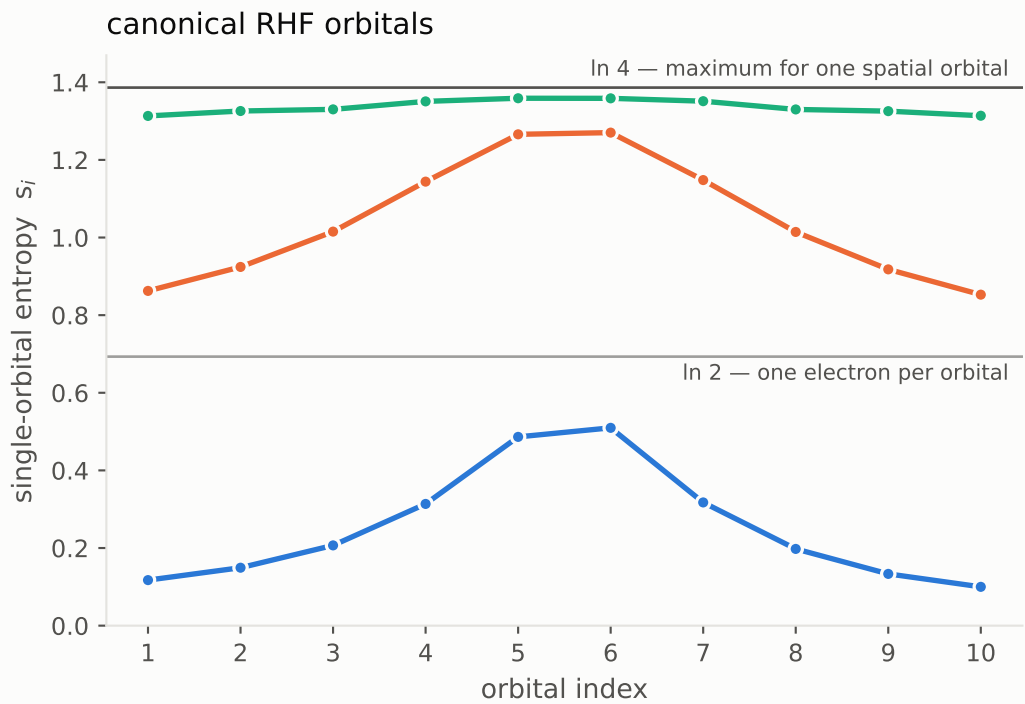


H₁₀/STO-3G single-orbital entropy — the same states in two orbital bases

The same three states in both panels, at identical energy — an orbital rotation cannot move it. Mean s_i runs $0.25 \rightarrow 1.34$ in canonical orbitals and $1.35 \rightarrow 0.74$ in localised ones: the bases swap roles, crossing near $R = 1.8 \text{ \AA}$. Near- $\ln 4$ values appear in whichever basis is wrong for the regime — delocalised orbitals for a stretched Mott-like chain, localised orbitals for a compressed metallic one. Saturation therefore diagnoses an unsuitable orbital partition, not a state beyond the method: DMRG reproduces exact FCI to 4×10^{-11} Ha at every point shown, with no bond-dimension truncation at all.



●— $R = 1 \text{ \AA}$ — weakly correlated ●— $R = 1.8 \text{ \AA}$ — moderately correlated ●— $R = 2.6 \text{ \AA}$ — strongly correlated